

Answer Key — Biology Grade 11

Sub-Strand 1.1: Taxonomy I • Lesson 4: Hierarchy of Classification: Climbing the Taxonomic Staircase

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6
B	A	B	A	D	B

USE AFTER: OBSERVE PHASE — Part A: Video 2 note-catcher on the order of the seven levels

Q1. Which list shows the seven main taxonomic levels in the correct order, from the broadest group to the most specific group?

- A. Species, Genus, Family, Order, Class, Phylum, Kingdom
- B. Kingdom, Phylum, Class, Order, Family, Genus, Species**
- C. Kingdom, Phylum, Class, Order, Genus, Family, Species
- D. Kingdom, Class, Phylum, Order, Family, Genus, Species

Correct answer: B. The hierarchy runs Kingdom, Phylum, Class, Order, Family, Genus, Species, from the broadest group to the most specific. The other lists swap levels or run in reverse.

USE AFTER: OBSERVE PHASE — Part B: taxonomic staircase card sorting (how groups change going down the hierarchy)

Q2. A learner moves down the taxonomic staircase from Family to Genus. What happens to the group of organisms?

- A. The group has fewer kinds of organisms, and they are more alike**
- B. The group has more kinds of organisms, and they are less alike
- C. The group has fewer kinds of organisms, but they are less alike
- D. The group has the same number of organisms, but they are named differently

Correct answer: A. Each step down the hierarchy is nested inside the one above. So it holds fewer kinds of organisms that share more features.

USE AFTER: EXPLAIN PHASE — Jigsaw: mixed groups compare the goat and chicken staircases

Q3. A goat is classified as Kingdom Animalia, Phylum Chordata, Class Mammalia. A domestic chicken is classified as Kingdom Animalia, Phylum Chordata, Class Aves. At which level do the goat and the chicken first separate into different groups?

- A. Species
- B. Class**
- C. Phylum
- D. Kingdom

Correct answer: B. The goat and chicken share Kingdom Animalia and Phylum Chordata. They first differ at Class, where the goat is Mammalia and the chicken is Aves.

USE AFTER: OBSERVE PHASE — Part B: filling in each level of the assigned organism (goat example) using the chart

Q4. In the classification of a goat (*Capra hircus*), the name Bovidae is given. At which taxonomic level does the name Bovidae belong?

- A. Family**
- B. Order
- C. Class
- D. Genus

Correct answer: A. For the goat, Artiodactyla is the Order and Bovidae is the Family. Capra is the Genus and Capra hircus is the Species.

USE AFTER: EXPLAIN PHASE — Jigsaw and CER: addressing misconceptions such as 'Species is the biggest group'

Q5. Which statement about the Species level is correct?

- A. Species is the second level, coming straight after Kingdom
- B. Species and Genus are the same level, so they mean the same thing
- C. Species is the biggest group, because it contains all the other levels
- D. Species is the most specific level, with the smallest group of the most similar organisms**

Correct answer: D. Species is the last and narrowest step of the staircase. It holds the smallest group of organisms that are most alike, and Genus is a separate, broader level above it.

USE AFTER: MODEL BUILDING PHASE — Model revision: marking the gap 'We still need one unique name for each species'

Q6. A farmer in Kericho and a student in Kisumu both use the taxonomic staircase from Kingdom to Species to classify a goat. Which problem is solved, and which problem still remains?

- A. Solved: all local names in Kikuyu, Luo and Kiswahili are now the same. Remaining: goats cannot be placed in a Kingdom.
- B. Solved: both can place the goat in the same position at every level. Remaining: we still need one unique name for each species that is recognised worldwide.**
- C. Solved: every organism now has one unique worldwide name. Remaining: people cannot agree on the order of the seven levels.
- D. Solved: organisms are grouped by colour and size in the same way everywhere. Remaining: nothing, because the system is complete.

Correct answer: B. The ranked hierarchy lets everyone put the goat in the same place at each level. The staircase does not yet give each species one agreed name, which leads to the next lesson.